STUDENT IMPLEMENTATION GUIDE

SEO and Publishing

A practical guide to improving and publishing a small HTML bakery website

What you will learn

- Improve page titles, descriptions, headings, images, links, and mobile layout.
- Create robots.txt and sitemap.xml to guide crawlers and help them discover pages.
- Improve image loading and follow basic security and privacy habits.
- Publish the website with GitHub Pages and check the live result.

Before you begin

Use your own website name, products, location, and contact details. Change the examples to match your project. SEO helps people and search engines understand a website; it cannot guarantee a particular ranking or that a page will be indexed.

Suggested project structure

```
MyBakeryWebsite/
├── index.html
├── about.html  services.html
├── enquiry.html  contact.html
├── robots.txt  sitemap.xml
├── css/style.css  js/script.js
└── images/
```

1 On-page SEO

On-page SEO improves the content and HTML on your own pages so visitors and search engines can understand them. Apply the steps to every page.

Choose a useful search phrase

Think of what a real customer might search. Choose a phrase that truthfully describes the page and business. Examples: home page — “bakery in Polokwane”; products — “cakes and cupcakes in Polokwane”; contact — “Sweet Treat Bakery contact details”. Use phrases naturally, not repeatedly.

Write a unique title and description

Put these inside each page’s <head>. Give every page its own accurate title and short summary. The title labels the browser tab and may appear as a search-result heading. The description summarizes the page and may be used as the result snippet.

```
<title>Bakery Products | Sweet Treat Bakery, Polokwane</title>
<meta name="description" content="Browse cakes, cupcakes and fresh bread from Sweet Treat Bakery in
Polokwane.">
```

Describe the home page in index.html, products in services.html, and the specific content of about.html, enquiry.html, and contact.html. Avoid copying one description across every page.

Structure headings

Use one meaningful <h1> for the main page subject, <h2> for major sections, and <h3> for subsections. Choose heading levels for meaning, not just text size.

```
<h1>Freshly Baked Products in Polokwane</h1>
<section>
  <h2>Our Cakes</h2>
  <h3>Birthday Cakes</h3>
</section>
```

Describe images and links

Use useful image filenames. Write alt text that describes the image for people who cannot see it. Keep it natural and specific.

```
<img src="images/chocolate-birthday-cake.jpg"
     alt="Chocolate birthday cake with vanilla icing">
```

Link text should tell visitors where a link goes. Avoid vague text such as “Click here” or “More”.

```
<a href="services.html">Browse our cakes, cupcakes and bread</a>
<a href="enquiry.html">Enquire about a custom cake</a>
```

Check mobile layout

Keep the viewport tag, resize the browser or use mobile preview, and check that navigation, product cards, images, and forms fit without sideways scrolling.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

2 Off-page SEO

Off-page SEO means actions outside your website that help people discover and trust the business. The goal is to build a real reputation and earn attention, not to manufacture ranking signals.

Earn relevant backlinks

A backlink is a link from another website to yours. Useful links may come from a local community organization, a school project showcase, a supplier directory, a market event page, or a local news story when the link is genuinely relevant.

1. Create something useful to share, such as a clear menu, seasonal product page, or bakery event announcement.
2. Tell relevant local organizations or event organizers about it when it helps their audience.
3. Ask partners to link to the bakery page that gives their visitors useful information.
4. Use normal, descriptive link text; do not ask sites to stuff links with repeated keywords.

Do not buy backlinks, use automated link-building services, or join schemes that swap links just to manipulate search rankings. Search engines treat those practices as link spam and they may harm visibility. A natural link is a genuine editorial choice by another site.

Promote the website on social media

Share the site's useful pages through the bakery's real social accounts. For example, post a seasonal menu and link to the Products page, or share opening-hour updates and link to Contact. Social posts help real people discover the website; do not promise students that likes or shares directly raise search rankings.

Improve local presence

For a real local business, keep its name, address or service area, phone number, hours, and website URL accurate and consistent wherever the business is listed. Add clear location and contact details to the website. A Google Business Profile is only appropriate for eligible businesses that meet Google's in-person customer-contact rules; a fictional classroom bakery should not create a profile pretending to be a real business.

Ask for honest reviews

A real business can invite genuine customers to leave honest reviews and respond politely to feedback. Do not write fake reviews, pay for reviews, or offer a reward in exchange for a positive review. Reviews must reflect genuine customer experiences.

Examples of off-page SEO

- Earn a local backlink: a community event page links to the bakery because it is supplying cakes for the event.
- Get listed by a relevant local organization: a market, business association, or useful local directory lists the bakery's correct website address.
- Share useful pages on social media: post a seasonal menu and link directly to the Products page. Social posts can bring visitors; do not promise that likes or shares directly improve rankings.
- Keep local business details consistent: use the same accurate business name, address or service area, phone number, opening hours, and website address on the website and eligible profiles.
- Invite genuine reviews: ask real customers to share their experience and respond politely. Do not buy reviews, write fake ones, or offer rewards for positive reviews.
- Earn local coverage: a community newspaper or food blog features the bakery because its story or products are relevant, and links to the site.

The shared idea is that other people mention the business because it is useful or relevant. Do not buy backlinks or join link-exchange schemes just to manipulate search rankings.

Student activity

For the bakery scenario, list three real or plausible local organizations that might share the website because it is useful to their audience. Draft one polite outreach message. Then identify one social post that would help customers find a relevant page. Do not claim that outreach guarantees backlinks or search placement.

Official guidance

- Google Search Essentials — https://developers.google.com/search/docs/essentials
- Google Search spam policies and link spam — https://developers.google.com/search/docs/essentials/spam-policies
- Google Business Profile eligibility — https://support.google.com/business/answer/13763036
- Google Business Profile representation guidelines — https://support.google.com/business/answer/3038177
- Google Business Profile prohibited and restricted content — https://support.google.com/business/answer/7400114

3 Robots, sitemap, speed, security

Create robots.txt

Create a plain-text file named robots.txt beside index.html. It gives crawlers instructions and points them to the sitemap. This file is public; never use it to hide private information.

```
User-agent: *
Allow: /

Sitemap: https://YOUR-PUBLISHED-SITE.example/sitemap.xml
```

Replace the example address after publishing. A small public site can usually allow crawling. Only add Disallow rules for a specific, appropriate reason.

Create sitemap.xml

The sitemap lists public pages you want crawlers to discover. Save it beside index.html and replace the example domain with the real published address.

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  <url><loc>https://YOUR-SITE.example/</loc></url>
  <url><loc>https://YOUR-SITE.example/about.html</loc></url>
  <url><loc>https://YOUR-SITE.example/services.html</loc></url>
  <url><loc>https://YOUR-SITE.example/enquiry.html</loc></url>
  <url><loc>https://YOUR-SITE.example/contact.html</loc></url>
</urlset>
```

A robots file and sitemap guide crawlers; they do not guarantee indexing or a search ranking.

Improve speed

Large images often slow pages. Resize images to a sensible display size and compress them. Use a suitable format such as WebP where supported. Lazy-load images farther down the page, but usually not the main image at the top.

```
<img src="images/bread.jpg"
     alt="Freshly baked loaves of bread"
     loading="lazy">
```

Security and privacy basics

- Never publish passwords, API keys, private data, or real customer details in code or the repository.
- Use the host's HTTPS option to serve the site over an encrypted connection.
- When opening an external link in a new tab, add rel="noopener noreferrer".
- Client-side form validation does not securely store or deliver submissions. Real form handling needs a trusted service or server-side code.

```
<a href="https://example.com" target="_blank"
   rel="noopener noreferrer">Visit our supplier</a>
```

4 Publish with GitHub Pages

GitHub Pages publishes static HTML, CSS, JavaScript, and images. It suits this assignment because the project already uses GitHub. With GitHub Free, the source repository must be public. Remove private information before publishing.

Prepare your repository

5. Make sure index.html is at the top level of the folder you will publish.
6. Include all HTML pages, CSS, JavaScript, images, robots.txt, and sitemap.xml.
7. Check filenames and capitalization in links and image paths. cake.jpg and Cake.jpg may be treated as different names.
8. Check that there are no secrets or customer details in the repository.
9. Commit and push the latest files to GitHub.

Enable GitHub Pages

10. Open the repository on GitHub and select Settings.
11. Select Pages in the sidebar.
12. Under Build and deployment, choose Deploy from a branch.
13. Choose the branch with the site, usually main.
14. Choose /(root) if index.html is at the top level. Otherwise choose the folder containing it.
15. Select Save and wait for deployment.
16. Open the published URL shown in the Pages settings.

A project URL commonly looks like https://GITHUB-USERNAME.github.io/REPOSITORY-NAME/

Check and update the live website

- Open the live URL and test each navigation link.
- Check that pages, images, styles, and JavaScript features load.
- Test the layout at a narrow/mobile width.
- If a file is missing, check its path, spelling, and capitalization.
- For later updates, commit and push again; after deployment, refresh and check the change.
- Replace the placeholder domain in robots.txt and sitemap.xml with the real site address.

Another free option: Netlify

Netlify can import a GitHub repository and redeploy when changes are pushed. For a plain HTML site, select the repository and branch, leave the build command empty, and choose the folder containing index.html as the publish directory. Free-plan limits can change, so check current terms.

5 Submission checklist

Use this list before submitting your repository and live-site link.

- ☐ Every page has a unique, accurate title and description.
- ☐ Pages have a clear main heading and organized section headings.
- ☐ Images have useful alt text and sensible file sizes.
- ☐ Links explain their destinations; related pages link to one another.
- ☐ The mobile layout fits without sideways scrolling.
- ☐ robots.txt is beside index.html and has the real sitemap address.
- ☐ sitemap.xml lists correct deployed URLs for public pages.
- ☐ No passwords, keys, private information, or customer data are published.
- ☐ The live site works and its pages, images, links, and interactions were checked.
- ☐ README includes the description, changelog, references, repository URL, and live URL.

Official references

- GitHub Pages — https://docs.github.com/en/pages/getting-started-with-github-pages/creating-a-github-pages-site
- Netlify deployment — https://docs.netlify.com/start/quickstarts/deploy-from-repository/
- Sitemaps protocol — https://www.sitemaps.org/protocol.html
- Google Search Central, robots.txt — https://developers.google.com/search/docs/crawling-indexing/robots/intro
- Google Search Central, SEO Starter Guide — https://developers.google.com/search/docs/fundamentals/seo-starter-guide